

ELEC5305 Project Feedback 2 Progress Report

Label Efficient Recognition of Traditional Chinese Instruments

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GitHub Project Link: <https://github.com/yyan0193/elec5305-project-550405308>

GitHub Pages Link: <https://yyan0193.github.io/elec5305-project-550405308/>

1 Stage 2 Overview

1.1 Outcomes

The second phase completed the project's core experimental workflow. Following feedback from the professor, I revised my work: the original instrument-recognition proposal evolved into a repeatable comparison of classical timbre features and a frozen music representation model under controlled labelling budgets. Specifically, this work examines how the quantity of labelled source audio affects traditional Chinese instrument recognition, and compares labelled data requirements between frozen MERT-v1-95M representations and compact, interpretable classical timbre representations. Current results show that MERT-v1-95M achieves higher segment-level macro-F1 across all tested labelling budgets, suggesting greater label efficiency, whereas the classical representation remains significantly more compact, faster, and more interpretable in terms of acoustic characteristics.

1.2 Work Completed in this Stage

- Audited the complete ChMusic dataset and created a traceable source-aware segment manifest.
- Reproduced the published MFCC and KNN benchmark as a pipeline validation step.
- Implemented a 64-dimensional classical timbre representation and a 768-dimensional frozen MERT representation.
- Ran source-aware five-fold experiments at four original-recording label budgets using the same classifier for both representations.
- Generated learning curves, confusion matrices, per-segment predictions, targeted error analysis, and computational-cost measurements.

2 Dataset Preparation and Leakage Control

The official ChMusic collection contains 55 original recordings from 11 traditional Chinese instruments, with five recordings per class. Each source recording was divided into

non-overlapping five-second segments, producing 988 segments. The original recording, rather than the segment, is treated as the independent data source.

A recording identifier is retained for every segment, and all segments from one source remain in the same fold. This prevents the model from being trained on one excerpt and tested on another excerpt from the same performance, recording room, microphone, or musical passage.

Dataset item	Stage 2 value
Instrument classes	11
Original recordings	55
Segment duration	5 seconds
Non-overlapping segments	988
Analysis sample rate	24 kHz

3 Implemented Methods

3.1 MFCC and KNN Benchmark Reproduction

The published ChMusic MFCC-KNN baseline was reproduced first. MFCC converts short audio frames into compact feature vectors that describe the spectral envelope and timbre of the sound. KNN then predicts an instrument by examining the majority class among its 15 nearest labelled examples. The implementation uses 20 MFCC coefficients and $K=15$, with recordings 1–4 used for training and recording 5 used for testing. Frame predictions are combined via majority voting to obtain a decision for each five-second segment. This benchmark validates the data and feature pipeline; it is not the primary classifier used in the controlled representation comparison.

3.2 Classical Timbre Representation

The classical representation comprises 64 features. Frame-level descriptors are summarised by their mean and standard deviation within each five-second segment. The feature vector includes statistics from 20 MFCCs alongside spectral centroid, bandwidth, roll-off, seven spectral-contrast bands, RMS energy and zero-crossing rate. These features enable classification as well as acoustic interpretation of timbral attributes including brightness, spectral width, energy and noisiness.

3.3 Frozen MERT Representation

The modern representation adopts the pretrained MERT-v1-95M model as a frozen feature extractor. MERT is not fine-tuned on the ChMusic dataset. Final hidden states undergo temporal mean pooling to yield a single 768-dimensional embedding for each segment. This setup

evaluates whether music-domain pretraining improves label efficiency, without additional neural network training on this small project dataset.

Representation	Dimension	Main content	Role in the project
Classical	64	MFCC and interpretable timbral statistics	Compact and acoustically explainable baseline
MERT-v1-95M	768	Frozen pretrained music embedding	Modern representation for label-efficiency comparison

4 Controlled Evaluation Design

Both representations are evaluated with the same StandardScaler and logistic-regression classifier. For each of five folds, recording number 1, 2, 3, 4, or 5 is held out for every instrument. Training uses all combinations of the remaining recordings at label budgets of one to four original recordings per class.

Label budget	Combinations per fold	Runs per representation
1	4	20
2	6	30
3	4	20
4	1	5

This produces 75 runs for each representation and 150 controlled runs in total. Recording-level macro-F1 is the primary metric, while segment-level macro-F1 and accuracy expose errors that can be hidden by recording-level probability aggregation. Training weights are balanced by source recording so that long recordings do not dominate the classifier merely because they generate more segments.

5 Interim Results

5.1 Benchmark reproduction

The reproduced MFCC and KNN segment accuracy is 94.152%, matching the published reference value of 94.15% to two decimal places. Segment macro-F1 is 0.9379 and recording-level accuracy is 1.0000. This confirms that the implemented preprocessing and baseline classification pipeline behave as expected.

5.2 Label efficiency

Labelled recordings per instrument	Classical segment macro-F1	MERT segment macro-F1
1	0.8718	0.9249
2	0.9198	0.9635
3	0.9393	0.9773
4	0.9547	0.9822

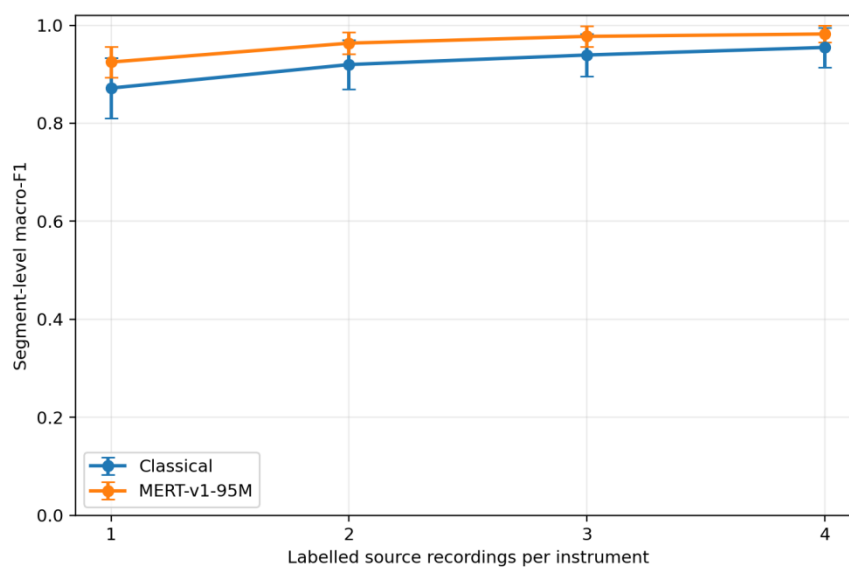


Figure 1 Segment-level macro-F1 across source-recording label budgets

MERT produces higher segment-level macro-F1 at every label budget. Its largest practical value appears in the low-label condition: with one labelled recording per instrument, MERT reaches 0.9249 macro-F1 compared with 0.8718 for the classical representation. At recording level, the two methods reach 1.0000 macro-F1 from three labelled recordings per class, indicating that recording aggregation is easier and can conceal remaining segment-level errors.

6 Error Analysis and Engineering Trade-Off

The initial confusion analysis shows that the pretrained representation does not improve every instrument pair equally. For Sheng and Dizi, the classical representation produces 14 symmetric five-second errors, whereas MERT produces none. For Yangqin and Guzheng, however, the classical representation produces two errors and MERT produces five. These contrasting cases will guide the final acoustic analysis using spectrograms and descriptor distributions.

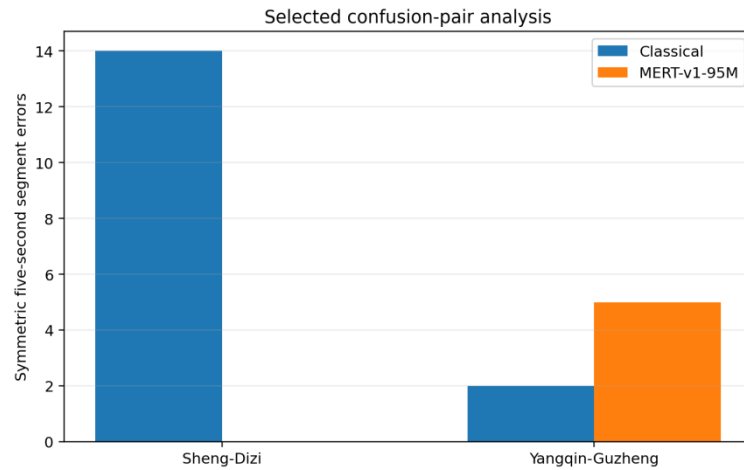


Figure 2 Selected symmetric segment errors in the full-recording evaluation

Representation	Dimension	Extraction time	Time per segment	Relative time
Classical	64	44.2 s	0.0447 s	1.0x
MERT-v1-95M	768	353.6 s	0.3579 s	8.0x

MERT improves low-label performance but uses a representation that is 12 times larger and requires approximately eight times more CPU feature-extraction time. The final conclusion should therefore describe a trade-off among predictive performance, annotation efficiency, computation, and interpretability rather than treating accuracy as the only criterion.

7 Stage 2 deliverables and validation

The repository now contains reproducible data-preparation, feature-extraction, benchmark, evaluation, error-analysis, and timing scripts; experiment configuration; metadata manifests; prediction files; result tables; and generated figures. Five automated tests pass, covering the expected dataset structure, source-aware separation, label-budget combinations, and recording-equal weighting.

8 Next stage

- Explain representative success and failure cases using spectrograms, MFCCs, and interpretable timbre descriptors.
- Add uncertainty and sensitivity checks without expanding the project into a multi-model leaderboard.

- Complete the final research report, GitHub documentation, demonstration program, and project video.
- State limitations carefully: the dataset has 988 segments but only 55 independent source recordings, and current findings should not be interpreted as universal instrument understanding.

Project resources

ChMusic dataset and baseline: <https://github.com/HaoranWeiUTD/ChMusic>

MERT-v1-95M model: <https://huggingface.co/m-a-p/MERT-v1-95M>